

AMIRMAHDI GHASABIOROJI

London, UK • Ghasabioroji@gmail.com • [linkedin.com/in/amirmahdi-ghasabioroji](https://www.linkedin.com/in/amirmahdi-ghasabioroji) • github.com/Amirmahdi-Ghasabioroji

EDUCATION

BSc Computer Science & Artificial Intelligence | Queen Mary, University of London *Sep 2024 – Present*
Main modules: NLP, Generative Deep Learning, Data systems, Probability, Big data processing, Bayesian decision making
Advanced Level Foundation: Mathematics: A*A*A equivalent | King's Education *Sep2023 – Jun 2024*

PROJECTS

Multi-Agent financial AI system / *Python, RAG, Qdrant, Docker, LangGraph, Cursor*

- ▶ Built quantitative strategy evaluation from scratch (Sharpe, Sortino, Calmar, drawdown, profit factor) across 8 rule-based trading playbooks, ranked via a transparent composite scoring system combining 3 weighted factors.
- ▶ Stress-tested strategy performance using a dual-layer simulation engine combining historical backtesting with a 5,000-path Monte Carlo forward bootstrap.
- ▶ Architected a 4-agent AI pipeline (Analyst, Risk, Strategy, Execution) orchestrated with LangGraph, combining RAG-based financial research (Qdrant vector DB, local Ollama LLM) with live risk analytics.
- ▶ Validated system reliability with 126+ automated unit and integration tests (pytest, Vitest) covering agent logic, simulation metrics, and API endpoints.

Bitcoin Market, Generative AI data visualisation / *Python, TensorFlow, Keras, NumPy, Plotly*

- ▶ Applied PCA and VAE for dimensionality reduction and unsupervised feature learning on live Binance and Bluesky data, achieving sub-500ms end-to-end processing.
- ▶ Built a custom MLP in TensorFlow for sentiment analysis and correlation visualisation, optimising loss functions to achieve error rate under 0.01.
- ▶ Validated model performance using quantitative metrics such as silhouette score, reconstruction accuracy, and entropy metrics.

NLP, Generative & Transformer Model Research / *PyTorch, TensorFlow*

- ▶ Designed and implemented autoregressive sequence models and Transformer-based architectures for text prediction, multimodal generation, and reward optimisation using RLHF.
- ▶ Built and refined training pipelines in PyTorch and TensorFlow with CUDA GPU acceleration, optimising loss functions and evaluation procedures to improve accuracy and generalisation while reducing training time, epochs and computing costs.

Horse Race Simulator / *Java, GitHub*

- ▶ Built a modular probabilistic simulation in Java with an interactive GUI and performance analytics tracking, using GitHub version control and OOP architecture to model financial outcomes under uncertainty.

EXPERIENCE

FX Specialist : Proprietary & Client Trading | Independent *May 2022 – Present*

- ▶ Developed and applied quantitative models and algorithmic trading strategies in Python, achieving a 15% increase in trade success rates through data-driven analysis and iterative refinement.
- ▶ Synthesised large volumes of time-series financial data into structured, actionable signals, applying statistical methods to inform position sizing and risk exposure.

Front of House Supervisor | Corretto, London *Jun 2025 – Present*

- ▶ Improved customer satisfaction rate and company performance by 15% through effective communication and leadership.

PASS Mentor & Logic Demonstrator | Queen Mary, University of London *Sep 2025 – Jun 2026*

- ▶ Raised course marks by 12% through structured peer-learning sessions; boosted academic engagement by up to 15% by translating complex mathematical and algorithmic concepts into accessible content.
- ▶ Collaborated with 10 demonstrators and lecturers to deliver structured technical sessions for the Logic & Discrete Structures module, strengthening cross-functional communication skills.

TECHNICAL SKILLS

AI / ML: Numpy, TensorFlow, Pytorch, Pandas, Scikit-learn, LangChain, agentic AI, generative models, supervised learning, unsupervised learning, fine-tuning, prompt engineering, Cursor, Docker

Data: Statistical modelling, dimensionality reduction, clustering, regression, data pipelines, financial modelling, algorithmic trading, quantitative analysis, Excel, SQL/MySQL